

nexMart Data Intelligence - Secure Terminal

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NEXMART DATA INTELLIGENCE
```

```
SECURE TERMINAL v0.4.1
```

```
[SYS] Establishing encrypted channel...
```

```
[SYS] Identity verified.
```

```
[SYS] Clearance level: CANDIDATE
```

```
[SYS] Loading mission brief...
```

```
[SYS] Done.
```

```
> Hello, agent.
```

```
You have been selected for evaluation by the Data Platform division.  
Congratulations.
```

```
The archive at BOOKS.TOSCRAPE.COM contains 1,000 catalogue entries  
spread across 50 paginated nodes. The data is exposed. Unguarded.  
But raw, inconsistent, and waiting to be structured.
```

```
Your task: extract it, clean it, store it. Then tell us what you  
noticed along the way – and how you would build around it.
```

```
We'll be watching the output. Not you. The output.
```

// MISSION PARAMETERS

[OBJECTIVE 01] - Extract

Write a Python script (`scraper.py`) that crawls the archive at **books.toscrape.com** and retrieves the following fields for **every entry across all 50 pages**:

- `title` - the book title
- `price` - price in GBP
- `rating` - star rating as an integer (1-5)
- `url` - direct link to the book's page

There are 1,000 records total. Your extractor must handle pagination.

The target is a practice site built specifically for scraping exercises. You are fully authorized to access it.

[OBJECTIVE 02] - Store

Write the extracted records to `output.csv`.

Requirements for the output file:

- Column names must be clean and consistent
- Price must be a number, no currency symbols
- Rating must be an integer, not a word (`3`, not `Three`)

[OBJECTIVE 03] - Debrief

Create `answers.md` with two sections. The second section contains two diagrams.

Observations - document 2 to 5 observations from building the scraper. A few sentences each. We are not looking for a report - we are looking for how you think. The more specific the better: reference actual field names, values, or decisions you made.

Things worth noting:

- Something in the raw data that wasn't in the format you expected (e.g. a specific field that was missing, malformed, or inconsistent)
- A trade-off in your implementation (e.g. why you chose one approach over another and what you gave up)

Design - create two diagrams using Mermaid (built into VS Code), draw.io, or any equivalent tool that supports entity-relationship and flow diagrams. Each diagram must be a separate file — either a `.md` file with a Mermaid code block, or a `.png` export.

- **Diagram 1 - ERD:** design a normalized relational schema for storing the scraped data. Explain your key decisions.
- **Diagram 2 - Data changes:** how would you design the system to detect and handle data changes over time - new entries appearing, prices changing, and entries being removed.

// DELIVERABLES

Compress the following into a zip file and return it through the channel you received this brief on:

```
archive/
├── scraper.py           ← your extraction script
├── output.csv          ← the data it produced
├── answers.md          ← debrief (objective 03)
├── diagram_erd.md/.png ← entity relation diagram
├── diagram_changes.md/.png ← data changes diagram
└── requirements.txt     ← all third-party libraries used
```

Estimated mission duration: 45 to 75 mins.

We are not looking for a polished production system. A clean, working solution with honest engineering reasoning is worth more than an over-engineered one that nobody can read.

// AUTHORIZED TOOLS

AI-assisted programming is allowed and encouraged (Claude, Copilot, ChatGPT, etc.).

This is standard operating procedure. Modern engineers use the tools available to them. We are evaluating your judgment, not your ability to write code without assistance.

What we will assess: do you understand what you built, can you explain your decisions, and did you bring your own thinking to the problem.

// KNOWN UNKNOWNNS - FAQ

```
> What is a Markdown (.md) file?
```

A Markdown file. Plain text with light formatting (`**bold**`, `- bullets`). Open it in any text editor. No special software needed.

```
> Which Python libraries am I allowed to use?
```

Any. Common choices: `requests` and `BeautifulSoup` for direct scraping, `Scrapy` if you prefer a framework. List everything in `requirements.txt`.

```
> Do I need Git or a repository?
```

No. Zip the folder. Send it back. That's it.

```
> What if I can't finish everything?
```

Submit what you have. Note in `answers.md` what you didn't get to and why. An honest incomplete submission is better than a rushed one.

```
> What if books.toscrape.com is unreachable?
```

If the site appears down, first confirm it is a global outage and not a local network issue: visit [isitdownrightnow - books.toscrape.com](https://isitdownrightnow.com/books.toscrape.com). If confirmed, notify us and wait. If it is not recovered after a few days, reach out asking for new instructions.

```
[SYS] Mission brief delivered.
```

```
[SYS] This terminal will not respond further.
```

```
[SYS] Good luck. You'll need it. Probably not, but it sounded right.
```

```
> _
```